

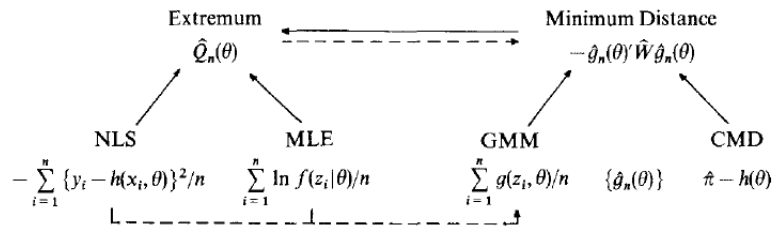
Estimation Principles

Raffaele Saggio

UBC

September 23, 2026

A Map of the World



How to estimate stuff?

- Common framework to analyzing many problems in econometrics summarized in Newey and McFadden (1994)
- Define quantity of interest θ_0 as maximizer of population criterion fn $Q_0(\theta)$
- Define estimator as maximizer of sample criterion fn $\hat{Q}_N(\theta)$
 - “Extremum” estimator (Amemiya, 1973)
 - θ of fixed dimension!
 - Need different tools when $\dim(\theta)$ grows with N

How to estimate stuff?

- Common framework to analyzing many problems in econometrics summarized in Newey and McFadden (1994)
- Define quantity of interest θ_0 as maximizer of population criterion fn $Q_0(\theta)$
- Define estimator as maximizer of sample criterion fn $\hat{Q}_N(\theta)$
 - “Extremum” estimator (Amemiya, 1973)
 - θ of fixed dimension!
 - Need different tools when $\dim(\theta)$ grows with N
- Basic results that hold subject to “usual” regularity conditions:
 - Consistency: $\hat{\theta} \xrightarrow{P} \theta_0$
 - Normality: $\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, H^{-1} V H^{-1})$

Preliminaries

The data are an *iid* sample $\{\mathbf{Z}_i\}_{i=1}^N$ from some d.f. $F_Z(\cdot)$

Parameter of interest is:

$$\theta_0 = \arg \max_{\theta \in \Theta} Q_0(\theta)$$

where $Q_0(\theta)$ is some population criterion function.

Assume θ_0 is a singleton (point-identification)

Guess your estimator!

Example 1?:

$$Q_0(\theta) = E \left[(Y_i - X_i' \theta)^2 \right]$$

Guess your estimator!

Example I?:

$$Q_0(\theta) = E \left[(Y_i - X_i' \theta)^2 \right]$$

Example II?:

$$Q_0(\theta) = E \left[Y_i \ln \Phi(X_i' \theta) + (1 - Y_i) \ln (1 - \Phi(X_i' \theta)) \right]$$

Guess your estimator!

Example I?:

$$Q_0(\theta) = E \left[(Y_i - X_i' \theta)^2 \right]$$

Example II?:

$$Q_0(\theta) = E \left[Y_i \ln \Phi(X_i' \theta) + (1 - Y_i) \ln (1 - \Phi(X_i' \theta)) \right]$$

Example III?:

$$Q_0(\theta) = E \left[(Y_i - X_i' \theta) Z_i \right]' E \left[Z_i Z_i' \right]^{-1} E \left[(Y_i - X_i' \theta) Z_i \right]$$

Estimator

Estimate θ by max'ing sample criterion fn:

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \hat{Q}_N(\theta)$$

Estimator

Estimate θ by max'ing sample criterion fn:

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \hat{Q}_N(\theta)$$

Example I:

$$\hat{Q}_N(\theta) = \frac{1}{N} \sum_{i=1}^N (Y_i - X_i' \theta)^2$$

Example II:

$$\hat{Q}_N(\theta) = \frac{1}{N} \sum_{i=1}^N Y_i \ln \Phi(X_i' \theta) + (1 - Y_i) \ln (1 - \Phi(X_i' \theta))$$

Example III:

$$\hat{Q}_N(\theta) = \left[\frac{1}{N} \sum_{i=1}^N (Y_i - X_i' \theta) Z_i \right]' \left[\frac{1}{N} \sum_{i=1}^N Z_i Z_i' \right]^{-1} \left[\frac{1}{N} \sum_{i=1}^N (Y_i - X_i' \theta) Z_i \right]$$

Consistency

When does $\hat{\theta} \xrightarrow{P} \theta_0$?

Intuition:

- Need for θ_0 to be “well separated”
- Need $\hat{Q}_N(\theta)$ “close” to $Q_0(\theta)$ for maximizer to be close

Standard sufficient conditions:

- $Q_0(\cdot)$ continuous
- Θ compact
- $\sup_{\theta \in \Theta} \left| \hat{Q}_N(\theta) - Q_0(\theta) \right| \xrightarrow{P} 0$ (uniform convergence)

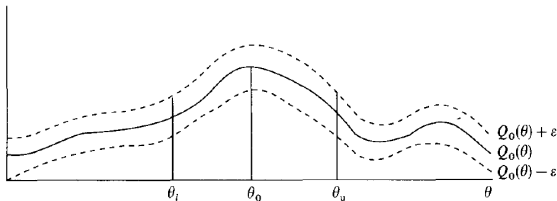
Uniform Convergence

Pointwise convergence usually easy to establish via LLN:

$$\left| \hat{Q}_N(\theta) - Q_0(\theta) \right| \xrightarrow{P} 0 \quad \forall \theta \in \Theta$$

Uniform convergence is stronger, need to trap “worst-case” deviation of fn from limit:

$$\sup_{\theta \in \Theta} \left| \hat{Q}_N(\theta) - Q_0(\theta) \right| \xrightarrow{P} 0$$



On finite grid, pointwise $\rightarrow$ uniform convergence. What can go wrong w/ continuous Θ ?

Theorem

If the following conditions hold:

- i) Θ is compact*
- ii) $Q_0(\theta)$ is uniquely maximized at θ_0*
- iii) $Q_0(\theta)$ is continuous*
- iv) $\hat{Q}_N(\theta)$ converges uniformly to $Q_0(\theta)$*

Then,

$$\hat{\theta} \xrightarrow{P} \theta_0$$

Proof

Asymptotic Distribution

Theorem

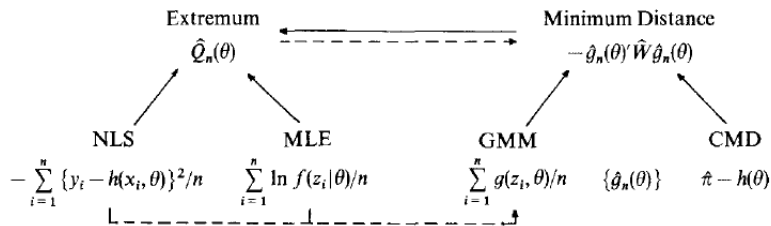
If $\hat{\theta} \xrightarrow{P} \theta_0$ and the following conditions hold:

- i) θ_0 is in the interior of Θ
- ii) $\hat{Q}_N(\theta)$ is twice differentiable in a neighborhood $\mathcal{N}$ of θ_0
- iii) $\sqrt{N} \nabla_{\theta} \hat{Q}_N(\theta_0) \xrightarrow{d} N(0, \mathbf{V})$
- iv) there is an $\mathbf{H}(\theta)$ that is continuous at θ_0 such that $\sup_{\theta} \left\| \nabla_{\theta\theta} \hat{Q}_N(\theta) - \mathbf{H}(\theta) \right\| \xrightarrow{P} 0$
- v) $\mathbf{H} \equiv \mathbf{H}(\theta_0)$ is nonsingular

Then,

$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \mathbf{H}^{-1} \mathbf{V} \mathbf{H}^{-1})$$

Special Cases



Classical Minimum Distance (CMD)

You have a model that says that certain observed moments in the data (collected all in a given vector $\boldsymbol{\pi}$) obeys a certain structure $\boldsymbol{g}(\theta_0)$.

$$\boldsymbol{\pi} - \boldsymbol{g}(\theta_0) = 0 \tag{1}$$

Classical Minimum Distance (CMD)

You have a model that says that certain observed moments in the data (collected all in a given vector $\boldsymbol{\pi}$) obeys a certain structure $\boldsymbol{g}(\theta_0)$.

$$\boldsymbol{\pi} - \boldsymbol{g}(\theta_0) = 0 \quad (1)$$

Objective fn (minimize instead of max):

$$\hat{Q}_N(\theta) = [\hat{\boldsymbol{\pi}} - \boldsymbol{g}(\theta)]' \hat{\boldsymbol{W}} [\hat{\boldsymbol{\pi}} - \boldsymbol{g}(\theta)]$$

- $\hat{\boldsymbol{\pi}} \xrightarrow{P} \boldsymbol{\pi}$ is a vector of reduced form sample moments
- $\boldsymbol{g}(\theta)$ is a structural function
- $\hat{\boldsymbol{W}} \xrightarrow{P} \boldsymbol{W}$ is a symmetric weighting matrix

Classical Minimum Distance (CMD)

You have a model that says that certain observed moments in the data (collected all in a given vector $\boldsymbol{\pi}$) obeys a certain structure $\boldsymbol{g}(\theta_0)$.

$$\boldsymbol{\pi} - \boldsymbol{g}(\theta_0) = 0 \quad (1)$$

Objective fn (minimize instead of max):

$$\hat{Q}_N(\theta) = [\hat{\boldsymbol{\pi}} - \boldsymbol{g}(\theta)]' \hat{\boldsymbol{W}} [\hat{\boldsymbol{\pi}} - \boldsymbol{g}(\theta)]$$

- $\hat{\boldsymbol{\pi}} \xrightarrow{P} \boldsymbol{\pi}$ is a vector of reduced form sample moments
- $\boldsymbol{g}(\theta)$ is a structural function
- $\hat{\boldsymbol{W}} \xrightarrow{P} \boldsymbol{W}$ is a symmetric weighting matrix

Classical Minimum Distance (CMD)

You have a model that says that certain observed moments in the data (collected all in a given vector $\boldsymbol{\pi}$) obeys a certain structure $\boldsymbol{g}(\theta_0)$.

$$\boldsymbol{\pi} - \boldsymbol{g}(\theta_0) = 0 \quad (1)$$

Objective fn (minimize instead of max):

$$\hat{Q}_N(\theta) = [\hat{\boldsymbol{\pi}} - \boldsymbol{g}(\theta)]' \hat{\boldsymbol{W}} [\hat{\boldsymbol{\pi}} - \boldsymbol{g}(\theta)]$$

- $\hat{\boldsymbol{\pi}} \xrightarrow{P} \boldsymbol{\pi}$ is a vector of reduced form sample moments
- $\boldsymbol{g}(\theta)$ is a structural function
- $\hat{\boldsymbol{W}} \xrightarrow{P} \boldsymbol{W}$ is a symmetric weighting matrix

Optimal weighting matrix: $\boldsymbol{W} = \text{Var}(\hat{\boldsymbol{\pi}})^{-1}$ (OMD)

Derivation of asymptotic variance

FOC:

$$\nabla_{\theta} \mathbf{g}(\hat{\theta})' \widehat{\mathbf{W}} [\hat{\pi} - \mathbf{g}(\hat{\theta})] = 0$$

Derivation of asymptotic variance

FOC:

$$\nabla_{\theta} \mathbf{g}(\hat{\theta})' \widehat{\mathbf{W}} [\hat{\pi} - \mathbf{g}(\hat{\theta})] = 0$$

Mean value expansion:

$$\mathbf{g}(\hat{\theta}) = \mathbf{g}(\theta_0) + \mathbf{G}(\bar{\theta})(\hat{\theta} - \theta_0)$$

where $\mathbf{G}(\theta) = \nabla_{\theta} \mathbf{g}(\theta)$. Substitute in to get:

$$\mathbf{G}(\hat{\theta})' \widehat{\mathbf{W}} [\hat{\pi} - \mathbf{g}(\theta_0) - \mathbf{G}(\bar{\theta})(\hat{\theta} - \theta_0)] = 0$$

Derivation of asymptotic variance

FOC:

$$\nabla_{\theta} \mathbf{g}(\hat{\theta})' \widehat{\mathbf{W}} [\hat{\pi} - \mathbf{g}(\hat{\theta})] = 0$$

Mean value expansion:

$$\mathbf{g}(\hat{\theta}) = \mathbf{g}(\theta_0) + \mathbf{G}(\bar{\theta})(\hat{\theta} - \theta_0)$$

where $\mathbf{G}(\theta) = \nabla_{\theta} \mathbf{g}(\theta)$. Substitute in to get:

$$\mathbf{G}(\hat{\theta})' \widehat{\mathbf{W}} [\hat{\pi} - \mathbf{g}(\theta_0) - \mathbf{G}(\bar{\theta})(\hat{\theta} - \theta_0)] = 0$$

Rearranging, normalizing, and taking limits yields:

$$\mathbf{G}(\theta_0)' \mathbf{W} \sqrt{N} [\hat{\pi} - \mathbf{g}(\theta_0)] = \mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0) \sqrt{N} (\hat{\theta} - \theta_0) + o_p(1)$$

Standard Errors

Solve for $\sqrt{N}(\hat{\theta} - \theta_0)$ to get:

$$\sqrt{N}(\hat{\theta} - \theta_0) = \left[\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0) \right]^{-1} \mathbf{G}(\theta_0)' \mathbf{W} \sqrt{N}[\hat{\pi} - \mathbf{g}(\theta_0)] + o_p(1)$$

By assumption:

$$\sqrt{N}[\hat{\pi} - \mathbf{g}(\theta_0)] \xrightarrow{d} N(0, \mathbf{V}_\pi)$$

Standard Errors

Solve for $\sqrt{N}(\hat{\theta} - \theta_0)$ to get:

$$\sqrt{N}(\hat{\theta} - \theta_0) = \left[\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0) \right]^{-1} \mathbf{G}(\theta_0)' \mathbf{W} \sqrt{N}[\hat{\pi} - \mathbf{g}(\theta_0)] + o_p(1)$$

By assumption:

$$\sqrt{N}[\hat{\pi} - \mathbf{g}(\theta_0)] \xrightarrow{d} N(0, \mathbf{V}_\pi)$$

Slutsky: $\sqrt{N}(\hat{\theta} - \theta_0)$ is (asymptotically) a linear combination of normals w/ variance taking usual sandwich form:

$$\left[\underbrace{\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0)}_H \right]^{-1} \underbrace{\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{V}_\pi \mathbf{W} \mathbf{G}(\theta_0)}_V \left[\underbrace{\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0)}_H \right]^{-1}$$

Standard Errors

Solve for $\sqrt{N}(\hat{\theta} - \theta_0)$ to get:

$$\sqrt{N}(\hat{\theta} - \theta_0) = \left[\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0) \right]^{-1} \mathbf{G}(\theta_0)' \mathbf{W} \sqrt{N}[\hat{\pi} - \mathbf{g}(\theta_0)] + o_p(1)$$

By assumption:

$$\sqrt{N}[\hat{\pi} - \mathbf{g}(\theta_0)] \xrightarrow{d} N(0, \mathbf{V}_\pi)$$

Slutsky: $\sqrt{N}(\hat{\theta} - \theta_0)$ is (asymptotically) a linear combination of normals w/ variance taking usual sandwich form:

$$\left[\underbrace{\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0)}_H \right]^{-1} \underbrace{\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{V}_\pi \mathbf{W} \mathbf{G}(\theta_0)}_V \left[\underbrace{\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0)}_H \right]^{-1}$$

Standard errors: replace unknown quantities with sample analogues (i.e. $\mathbf{G}(\hat{\theta})$ for $\mathbf{G}(\theta_0)$, $\hat{\mathbf{V}}_\pi$ for $\mathbf{V}_\pi$)

Optimal Weighting

With optimal weights $\mathbf{W} = \mathbf{V}_\pi^{-1}$, asymptotic variance reduces to:

$$\left[\mathbf{G}(\theta_0)' \mathbf{V}_\pi^{-1} \mathbf{G}(\theta_0) \right]^{-1}$$

Optimal Weighting

With optimal weights $\mathbf{W} = \mathbf{V}_\pi^{-1}$, asymptotic variance reduces to:

$$\left[\mathbf{G}(\theta_0)' \mathbf{V}_\pi^{-1} \mathbf{G}(\theta_0) \right]^{-1}$$

Specification testing: optimal weighting yields null distribution for minimized value of criterion function

$$\hat{Q}_N^{OMD}(\hat{\theta}) \sim \chi^2(J - K)$$

Answers question: does my model explain the data up to sampling error? (i.e. could population $R^2 = 1$?)

Minimum Distance is the Purest Form of Econometrics

- We are now going to analyze the paper by Card and Lemieux (2001, QJE) to better understand, via a real research question, how MD works in practice.

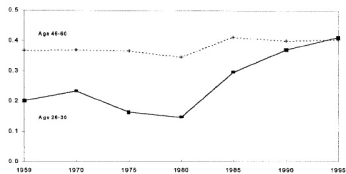
- Fact: Inequality increased a lot in the US: gap in wages for men b/w college and HS diploma was about 25 percent in the mid-1970s and grew to 40 percent in 1998.

- Fact: Inequality increased a lot in the US: gap in wages for men b/w college and HS diploma was about 25 percent in the mid-1970s and grew to 40 percent in 1998.
- By the time CL wrote their paper the standard theory to explain this fact is SBTC: arrival of computers benefits mostly highly skilled workers.

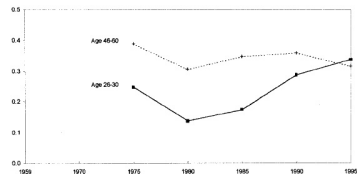
- Fact: Inequality increased a lot in the US: gap in wages for men b/w college and HS diploma was about 25 percent in the mid-1970s and grew to 40 percent in 1998.
- By the time CL wrote their paper the standard theory to explain this fact is SBTC: arrival of computers benefits mostly highly skilled workers.
- CL argue instead that a key part of the story is the decline in the supply of college-educated workers in most recent cohorts.

Starting observation

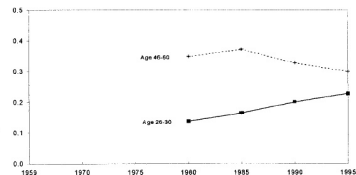
A. United States



B. United Kingdom



C. Canada



724

QUARTERLY JOURNAL OF ECONOMICS

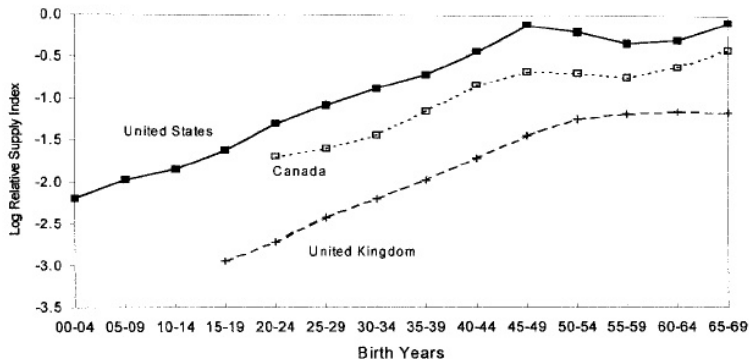


FIGURE IV
Relative Supply of College-Educated Workers by Cohort

Nested CES

- Aggregate production function

$$y_t = (\theta_{ht} H_t^\rho + \theta_{ct} C_t^\rho)^{\frac{1}{\rho}} \quad (2)$$

where (H_t, C_t) are high school and college equivalent labor aggregates of age groups j :

$$H_t = \left[\sum_j \left(\alpha_j H_{jt}^\eta \right) \right]^{\frac{1}{\eta}} \quad (3)$$

$$C_t = \left[\sum_j \left(\beta_j C_{jt}^\eta \right) \right]^{\frac{1}{\eta}} \quad (4)$$

- The relevant elasticities are

$$\sigma_A = (1 - \eta)^{-1} \quad \sigma_E = (1 - \rho)^{-1} \quad (5)$$

- If wages are set according to marginal product of labor, then

$$\log \frac{w_{jt}^c}{w_{jt}^h} = \log \frac{\theta_{ct}}{\theta_{ht}} + \log \frac{\beta_j}{\alpha_j} + (\rho - \eta) \log \frac{C_t}{H_t} + (\eta - 1) \log \frac{C_{jt}}{H_{jt}} \quad (6)$$

- What is the key assumption here?

Interpretation

- It is useful to rearrange the last expression as

$$\log \frac{w_{jt}^c}{w_{jt}^h} = \log \frac{\theta_{ct}}{\theta_{ht}} + \log \frac{\beta_j}{\alpha_j} - \frac{1}{\sigma_E} \log \frac{C_t}{H_t} - \frac{1}{\sigma_A} \left[\log \frac{C_{jt}}{H_{jt}} - \log \frac{C_t}{H_t} \right] \quad (7)$$

- How to model relative supplies?
 - Relative supply of college grads for people aged j largely determined by the college enrollment decision that these individuals made when they were 20

$$\log \frac{C_{jt}}{H_{jt}} = \underbrace{\lambda_{t-j}}_{\text{Cohort Effect}} + \underbrace{\phi_j}_{\text{Age}} \quad (8)$$

age effect is constant over-time!

- So we get that

$$\underbrace{\log \frac{w_{jt}^c}{w_{jt}^h}}_{r_{jt}} = \underbrace{\log \frac{\theta_{ct}}{\theta_{ht}}}_{\text{Year Effects}} + \underbrace{\log \frac{\beta_j}{\alpha_j} - \frac{1}{\sigma_A} \phi_j}_{\text{Age}} - \underbrace{\left(\frac{1}{\sigma_E} - \frac{1}{\sigma_A} \right) \log \frac{C_t}{H_t}}_{\text{Year Effects}} - \underbrace{\frac{1}{\sigma_A} \lambda_{t-j}}_{\text{Cohort}} \quad (9)$$

Decomposing the returns

- CL model implies the following *reduced form* when considering gap in wages between C and H across age groups and time interval

$$r_{jt} = b_j + c_{t-j} + d_t \quad (10)$$

- This is precisely a Minimum Distance framework...
- Data: Wage premiums for men between 1970 to 1997 for age groups 26 to 60 years old.
 - 1 What is the vector of moments, π ?
 - 2 What is the vector of (reduced form) parameters: θ ?
 - 3 How many moments do you have?
 - 4 How many (reduced-form) parameters do we have?
 - 5 What would be the optimal weighting matrix here?
 - 6 What would you do to test the model validity?

Do we pass the test?

TABLE II
DECOMPOSITIONS OF COLLEGE-HIGH SCHOOL WAGE DIFFERENTIALS BY AGE AND
YEAR INTO COHORT, AGE, AND TIME EFFECTS

	United States			United Kingdom			Canada		
	No cohort effects	10 oldest cohorts only	10 oldest coh. eff. same	No cohort effects	7 oldest cohorts only	7 oldest coh. eff. same	No cohort effects	6 oldest cohorts only	6 oldest coh. eff. same
Year effects									
1970	0.026 (0.021)	0.020 (0.010)	0.020 (0.009)	—	—	—	—	—	—
1975	-0.020 (0.021)	-0.024 (0.010)	-0.024 (0.009)	0.000	0.000	0.000	—	—	—
1980	-0.049 (0.019)	-0.060 (0.011)	-0.062 (0.009)	-0.077 (0.026)	-0.086 (0.021)	-0.076 (0.018)	0.000	0.000	0.000
1985	0.058 (0.020)	0.017 (0.013)	0.015 (0.010)	-0.045 (0.027)	-0.057 (0.025)	-0.069 (0.021)	0.020 (0.019)	0.007 (0.017)	-0.004 (0.013)
1990	0.099 (0.020)	0.022 (0.015)	0.022 (0.011)	0.021 (0.028)	-0.041 (0.028)	-0.037 (0.025)	0.031 (0.017)	-0.011 (0.018)	-0.025 (0.016)
1995	0.141 (0.021)	0.024 (0.019)	0.034 (0.014)	0.051 (0.030)	-0.060 (0.038)	-0.039 (0.031)	0.043 (0.017)	-0.038 (0.021)	-0.039 (0.021)
Cohort effects:									
1950-1954	—	—	0.040 (0.011)	—	—	-0.009 (0.019)	—	—	0.028 (0.015)
1955-1959	—	—	0.124 (0.013)	—	—	0.075 (0.025)	—	—	0.076 (0.021)
1960-1964	—	—	0.178 (0.016)	—	—	0.134 (0.032)	—	—	0.133 (0.027)
1964-1969	—	—	0.175 (0.024)	—	—	0.162 (0.046)	—	—	0.142 (0.036)
Degrees of freedom	36	26	32	24	14	20	18	9	14
χ^2 (p-value)	295.01 (0.00)	48.84 (0.00)	51.09 (0.02)	48.31 (0.00)	10.76 (0.70)	15.33 (0.76)	66.06 (0.00)	12.00 (0.21)	20.51 (0.11)
R^2	0.87	0.97	0.98	0.77	0.85	0.92	0.89	0.90	0.97

Standard errors are in parentheses. Models are fit by weighted least squares to the age-group by year college-high school wage gaps shown in Table I. Weights are inverse sampling variances of the estimated wage gaps. All models include age effects. For the United States and the United Kingdom, the years indicated when reporting the estimated year effects are the midpoints of the year intervals shown in Table I. The base (omitted) year is 1959 for the United States, 1975-1977 for the United Kingdom, and 1980 for Canada.

Practical Problems w/ OMD

Optimal weighting behaves poorly when $\mathbf{W}$ is large relative to sample size (Altonji and Segal, 1996)

- Variant of “Kakwani bias” in FGLS
- Problem emerges from correlation between weights and moments

Practical Problems w/ OMD

Optimal weighting behaves poorly when $\mathbf{W}$ is large relative to sample size (Altonji and Segal, 1996)

- Variant of “Kakwani bias” in FGLS
- Problem emerges from correlation between weights and moments

Potential solutions:

- Use inefficient (but tractable) $\mathbf{W}$: equal (or diagonal) weight matrix (e.g., Abowd and Card, 1992)
- Jackknife / split sample estimation of $\mathbf{W}$ (Kezdi, Solon, and Hahn, 2002)
- Parameterize $\mathbf{W} = \mathbf{W}(\delta)$ and estimate δ along with θ via “CUGMM” (Hansen, Heaton, and Yaron, 1996; Hausman et al, 2011)
- Generalized Empirical Likelihood (Newey and Smith, 2004)

Spec test without optimal weighting

- Is it possible to test our model if we don't have (want) to use optimal weighting matrix?

- Newey (1985): yes! Consider estimation of θ that is based upon

$$[\hat{\pi} - g(\theta)]' \mathbf{A} [\hat{\pi} - g(\theta)] \quad (11)$$

where $\mathbf{A}$ is a possibly stochastic matrix.

- Newey showed that

$$N[\hat{\pi} - g(\theta)]' \mathbf{R}^- [\hat{\pi} - g(\theta)] \sim \chi^2_{J-K} \quad (12)$$

where $\mathbf{R}^-$ is the generalized inverse of the matrix $\mathbf{R} = \mathbf{M} \mathbf{V}_\pi \mathbf{M}'$ with

$$\mathbf{M} = \mathbf{I} - \mathbf{G}(\hat{\theta})(\mathbf{G}(\hat{\theta})' \mathbf{A} \mathbf{G}(\hat{\theta}))^{-1} \mathbf{G}(\hat{\theta})' \mathbf{A} \quad (13)$$

- Need generalized inverse because $\mathbf{R}$ does not have full rank (presence of $\mathbf{M}$).

On the frontier

- Important assumption is “correct” specification.

$$\sqrt{N}(\hat{\pi} - g(\theta_0)) \sim^a \mathcal{N}(0, \mathbf{V}_\pi) \quad (14)$$

- Important assumption is “correct” specification.

$$\sqrt{N}(\hat{\pi} - g(\theta_0)) \sim^a \mathcal{N}(0, \mathbf{V}_\pi) \quad (14)$$

- What happens under misspecification?
- Chamberlain (1994): be careful with weights! Randomness of weights increases noise in resulting estimator.
- Active frontier in econometrics today: what if my model is wrong?
Can my standard errors account for that?
 - Armstrong and Kolesar (2018); Bonhomme and Weidner (2019); Armstrong-Kline-Sun (2023).

Criterion function (min once again):

$$\hat{Q}_{GMM}(\theta) = \hat{\mathbf{g}}(\theta)' \hat{\mathbf{W}} \hat{\mathbf{g}}(\theta)$$

- $\hat{\mathbf{g}}(\theta) = \frac{1}{N} \sum_i \mathbf{f}(\mathbf{Z}_i, \theta)$ is a $J \times 1$ vector of moment conditions
- CMD nested by separable case where $\mathbf{f}(\mathbf{Z}_i, \theta) = \boldsymbol{\pi}(\mathbf{Z}_i) - \mathbf{g}(\theta)$
- MM nested by just-id case where $J = \dim(\theta)$

Identification

Population moment restrictions:

$$E[\mathbf{f}(\mathbf{Z}_i, \theta)] = 0 \text{ iff } \theta = \theta_0$$

Examples:

- Instrumental variables (orthogonality condition):

$$E[(Y_i - X_i'\beta) Z_i] = 0$$

- Rational expectations restrictions (e.g., Hall, 1978) (real interest rate = 0)

$$u'(C_t) = \beta E[u'(C_{t+1}) | \Omega_t]$$

Derivation of Asymptotic Variance

FOC:

$$\widehat{\mathbf{G}}(\widehat{\theta})' \widehat{\mathbf{W}} \widehat{\mathbf{g}}(\widehat{\theta}) = 0$$

Mean value expansion:

$$\widehat{\mathbf{g}}(\widehat{\theta}) = \widehat{\mathbf{g}}(\theta_0) + \widehat{\mathbf{G}}(\bar{\theta})(\widehat{\theta} - \theta_0)$$

Substitute in to get:

$$\widehat{\mathbf{G}}(\widehat{\theta})' \widehat{\mathbf{W}} \widehat{\mathbf{g}}(\theta_0) + \widehat{\mathbf{G}}(\widehat{\theta})' \widehat{\mathbf{W}} \widehat{\mathbf{G}}(\bar{\theta})(\widehat{\theta} - \theta_0) = 0$$

Therefore

$$\sqrt{N}(\widehat{\theta} - \theta_0) = (\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0))^{-1} \mathbf{G}(\theta_0)' \mathbf{W} \sqrt{N} \widehat{\mathbf{g}}(\theta_0) + o_p(1)$$

Derivation of Asymptotic Variance

By assumption:

$$E \left[\sqrt{N} \hat{\mathbf{g}}(\theta_0) \right] = 0$$

Hence, for $\mathbf{V}_f = E \left[\mathbf{f}(\mathbf{Z}_i, \theta_0) \mathbf{f}(\mathbf{Z}_i, \theta_0)' \right]$, we have

$$AVAR \left(\mathbf{G}(\theta_0)' \mathbf{W} \sqrt{N} \hat{\mathbf{g}}(\theta_0) \right) = \mathbf{G}(\theta_0)' \mathbf{W} \mathbf{V}_f \mathbf{W} \mathbf{G}(\theta_0)$$

Consequently $AVAR \left(\sqrt{N} \left(\hat{\theta} - \theta_0 \right) \right)$ is:

$$\left(\underbrace{\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0)}_H \right)^{-1} \underbrace{\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{V}_f \mathbf{W} \mathbf{G}(\theta_0)}_V \left(\underbrace{\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0)}_H \right)^{-1}$$

Plugin sample analogues to get standard errors

Optimal Weighting

When $\mathbf{W} = \mathbf{V}_f^{-1}$ (optimal weighting) AVAR simplifies to:

$$(\mathbf{G}(\theta_0)' \mathbf{V}_f^{-1} \mathbf{G}(\theta_0))^{-1}$$

2-step estimation approach (ala 3SLS):

- 1 get inefficient starting estimates $\hat{\theta}^{(1)}$ from choosing $\hat{\mathbf{W}} = \mathbf{I}$.
- 2 minimize $Q_{GMM}(\theta)$ using $\hat{\mathbf{W}} = \mathbf{W}(\hat{\theta}^{(1)})$

Keep going?: further steps provide no (first order) asymptotic advantage but often perform better in finite samples

CUGMM (Hansen, Heaton, and Yaron, 1996)

As w/ OMD, problems emerge when weight matrix has too many unknown parameters

One solution: “continuously update” weight matrix $\mathbf{W}(\theta)$ and minimize

$$\hat{Q}_{CUGMM}(\theta) = \hat{\mathbf{g}}(\theta)' \hat{\mathbf{W}}(\theta) \hat{\mathbf{g}}(\theta)$$

Better asymptotic performance than GMM but can be difficult to find minimum.

Maximum Likelihood

You are willing to specify the *entire* conditional distribution of the outcome given covariates.

Maximum Likelihood

You are willing to specify the *entire* conditional distribution of the outcome given covariates. Objective fn:

$$\hat{Q}_{ML}(\theta) = \frac{1}{N} \sum_i l(\mathbf{z}_i, \theta)$$

FOC:

$$\frac{1}{N} \sum_i \mathbf{s}(\mathbf{z}_i, \hat{\theta}) = 0$$

- ML: if model is correct $\rightarrow E[\mathbf{s}(\mathbf{Z}_i, \theta_0)] = 0$
- No weighting problems!
- Likelihood chooses the right moments for you! (embedded in the score function)
 - “Efficient Method of Moments” on FOCs (Gallant and Tauchen, 1996)
- ML is prone to misspecification. Even getting a minor feature of the entire distribution wrong can make the ML estimator inconsistent.
- Can you think of a relevant counter-example?

Asymptotic variance

“Influence function” representation – 1st order effect of adding an obs on estimator:

$$\sqrt{N} \left(\hat{\theta} - \theta_0 \right) = \hat{\mathbf{H}}(\theta_0)^{-1} \frac{1}{\sqrt{N}} \sum_i \mathbf{s}(\mathbf{Z}_i, \theta_0) + o_p(1)$$

Variance of IF gives asymptotic variance:

$$\mathbf{H}(\theta_0)^{-1} \mathbf{V} \mathbf{H}(\theta_0)^{-1}$$

where $\mathbf{V} = \mathbf{E} \left[\mathbf{s}(\mathbf{Z}_i, \theta_0) \mathbf{s}(\mathbf{Z}_i, \theta_0)' \right]$

Asymptotic variance

“Influence function” representation – 1st order effect of adding an obs on estimator:

$$\sqrt{N}(\hat{\theta} - \theta_0) = \hat{H}(\theta_0)^{-1} \frac{1}{\sqrt{N}} \sum_i \mathbf{s}(\mathbf{Z}_i, \theta_0) + o_p(1)$$

Variance of IF gives asymptotic variance:

$$\mathbf{H}(\theta_0)^{-1} \mathbf{V} \mathbf{H}(\theta_0)^{-1}$$

where $\mathbf{V} = \mathbf{E} [\mathbf{s}(\mathbf{Z}_i, \theta_0) \mathbf{s}(\mathbf{Z}_i, \theta_0)']$

Information matrix equality:

$$\mathbf{V} = \mathbf{H}(\theta_0)$$

When this holds, asymptotic variance reduces to Cramer-Rao lower bound $\mathbf{H}(\theta_0)^{-1}$.

- Nothing beats the CRLB $\rightarrow$ nothing beats a properly specified likelihood model.

Simulation Estimators

Often difficult to write down the likelihood

- Particularly difficult for multinomial choice problems where regions of integration can quickly become intractable
- Or for parametric models with lots of unobserved heterogeneity

Simulation Estimators

Often difficult to write down the likelihood

- Particularly difficult for multinomial choice problems where regions of integration can quickly become intractable
- Or for parametric models with lots of unobserved heterogeneity

Simulation methods

- Can use computer to compute likelihood via simulation (MSL)
- With continuous variables often easier to simulate moment conditions (MSM)
- Great (free) introduction by Train (2009) or Chapter 12 of Cameron and Trivedi.

Simulation Estimators

Often difficult to write down the likelihood

- Particularly difficult for multinomial choice problems where regions of integration can quickly become intractable
- Or for parametric models with lots of unobserved heterogeneity

Simulation methods

- Can use computer to compute likelihood via simulation (MSL)
- With continuous variables often easier to simulate moment conditions (MSM)
- Great (free) introduction by Train (2009) or Chapter 12 of Cameron and Trivedi.

Warning: can easily waste a lot of time on this stuff

- Tendency to use these methods when you don't fully understand the model / identification
- Important to “warm up” by estimating simpler models that you understand!

Maximum Simulated Likelihood

$$\hat{\theta}_{MSL} \equiv \arg \max_{\theta} \frac{1}{N} \sum_{i=1}^N \ln \hat{L}_M(\mathbf{Z}_i, \theta)$$

- $\hat{L}_M(\mathbf{Z}_i, \theta) = \hat{L}(\mathbf{Z}_i, \theta, u_{i1}, u_{i2}, \dots, u_{iM})$ is simulated likelihood
- Fully parametric: drawing $\{u_{im}\}_{m=1}^M$ from known distribution
- Hold $\{\mathbf{u}_i\}_{i=1}^N$ fixed when searching over θ to avoid “chatter”
- Need large M to ensure consistency:
$$\lim_{M \rightarrow \infty} P(|\hat{L}_M(\mathbf{Z}_i, \theta) - L(\mathbf{Z}_i, \theta)| > \varepsilon) = 0$$

Example (Probit)

Probit likelihood:

$$L(Y_i, X_i, \beta) = \Phi(X_i' \beta)^{Y_i} [1 - \Phi(X_i' \beta)]^{1-Y_i}$$

Simulated analogue:

$$\begin{aligned} \hat{L}_M(Y_i, X_i, \beta) &= \left(\frac{1}{M} \sum_{m=1}^M 1[X_i' \beta - u_{im} > 0] \right)^{Y_i} \\ &\quad \times \left(\frac{1}{M} \sum_{m=1}^M 1[X_i' \beta - u_{im} < 0] \right)^{1-Y_i} \end{aligned}$$

where $\left\{ u_{im} \stackrel{iid}{\sim} N(0, 1) \right\}_{i=1, m=1}^{N, M}$

- Unbiased simulator: $E[\hat{L}_M(Y_i, X_i, \beta)] = L(Y_i, X_i, \beta)$
- But $\hat{L}_M(Y_i, X_i, \beta)$ nondifferentiable for finite M (step function)

Example (Random Coefficient Logit)

Suppose

$$Y_i = 1 [b_i X_i + \varepsilon_i > 0]$$

where $\varepsilon_i \sim \text{Logistic}(0, 1)$ and $b_i \sim N(\mu, \sigma^2)$

- Could simulate likelihood by taking draws from (ε_i, b_i) .
- Better to integrate out ε_i analytically to obtain smooth objective fn:

$$\begin{aligned} \hat{L}_M(Y_i, X_i, \mu, \sigma) &\equiv \left(\frac{1}{M} \sum_{m=1}^M \Lambda((\mu + \sigma u_{im}) X_i) \right)^{Y_i} \\ &\times \left(1 - \frac{1}{M} \sum_{m=1}^M \Lambda((\mu + \sigma u_{im}) X_i) \right)^{1-Y_i} \end{aligned}$$

Asymptotics

Pakes and Pollard (1989): general asymptotic framework for simulation estimators (“stochastic equicontinuity”)

Gourieroux and Monfort (1991): if $M, N \rightarrow \infty$ and $\sqrt{N}/M \rightarrow 0$, then $\hat{\theta}_{MSL} \xrightarrow{P} \theta_0$ and

$$\sqrt{N} \left(\hat{\theta}_{MSL} - \theta_0 \right) \xrightarrow{d} N \left(0, \mathbf{H}(\theta_0)^{-1} \mathbf{V} \mathbf{H}(\theta_0)^{-1} \right)$$

Asymptotics

Pakes and Pollard (1989): general asymptotic framework for simulation estimators (“stochastic equicontinuity”)

Gourieroux and Monfort (1991): if $M, N \rightarrow \infty$ and $\sqrt{N}/M \rightarrow 0$, then $\hat{\theta}_{MSL} \xrightarrow{P} \theta_0$ and

$$\sqrt{N} \left(\hat{\theta}_{MSL} - \theta_0 \right) \xrightarrow{d} N \left(0, \mathbf{H}(\theta_0)^{-1} \mathbf{V} \mathbf{H}(\theta_0)^{-1} \right)$$

Practical advice: start with $M = 100$ to get initial estimates then check stability as M increases

Asymptotics

Pakes and Pollard (1989): general asymptotic framework for simulation estimators (“stochastic equicontinuity”)

Gourieroux and Monfort (1991): if $M, N \rightarrow \infty$ and $\sqrt{N}/M \rightarrow 0$, then $\hat{\theta}_{MSL} \xrightarrow{P} \theta_0$ and

$$\sqrt{N} \left(\hat{\theta}_{MSL} - \theta_0 \right) \xrightarrow{d} N \left(0, \mathbf{H}(\theta_0)^{-1} \mathbf{V} \mathbf{H}(\theta_0)^{-1} \right)$$

Practical advice: start with $M = 100$ to get initial estimates then check stability as M increases

Std errors: plug in simulation estimators of $\mathbf{H}(\hat{\theta}_{MSL})$, $\mathbf{V}(\hat{\theta}_{MSL})$

Method of Simulated Moments

$$\hat{\theta}_{MSM} \equiv \arg \min_{\theta} \hat{\mathbf{g}}_M(\theta)' \hat{\mathbf{W}} \hat{\mathbf{g}}_M(\theta)$$

- $\hat{\mathbf{g}}_M(\theta) = \frac{1}{N} \sum_i \hat{\mathbf{f}}_M(\mathbf{Z}_i, \theta)$ is simulated moment condition
- McFadden (1989): for fixed M , as $N \rightarrow \infty$, $\hat{\theta}_{MSM} \xrightarrow{p} \theta_0$ and $AVAR\left(\sqrt{N}\left(\hat{\theta}_{MSM} - \theta_0\right)\right) =$

$$\left(\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0)\right)^{-1} \mathbf{G}(\theta_0)' \mathbf{W} \mathbf{V}_{f,M} \mathbf{W} \mathbf{G}(\theta_0) \left(\mathbf{G}(\theta_0)' \mathbf{W} \mathbf{G}(\theta_0)\right)^{-1}$$

Example

- Let's see one of these simulated tools in action using a concrete example: Duflo-Hanna-Ryan (2013,AER).
- Teacher absenteeism is a serious problem in many developing countries.
- In DHR setting: 40% absent rate
- DHR provided tamper-proof cameras to 60 randomly selected schools and instructed a student in the class to take pictures beginning and the end of the day with students.
- Teacher in treated schools also received a different compensation package

$$Y_{im} = 500 + 50 \times \mathbf{1}\{X_{im} > 10\} \times (X_{im} - 10) \quad (15)$$

- Control group: Flat 1000 Rs per month.

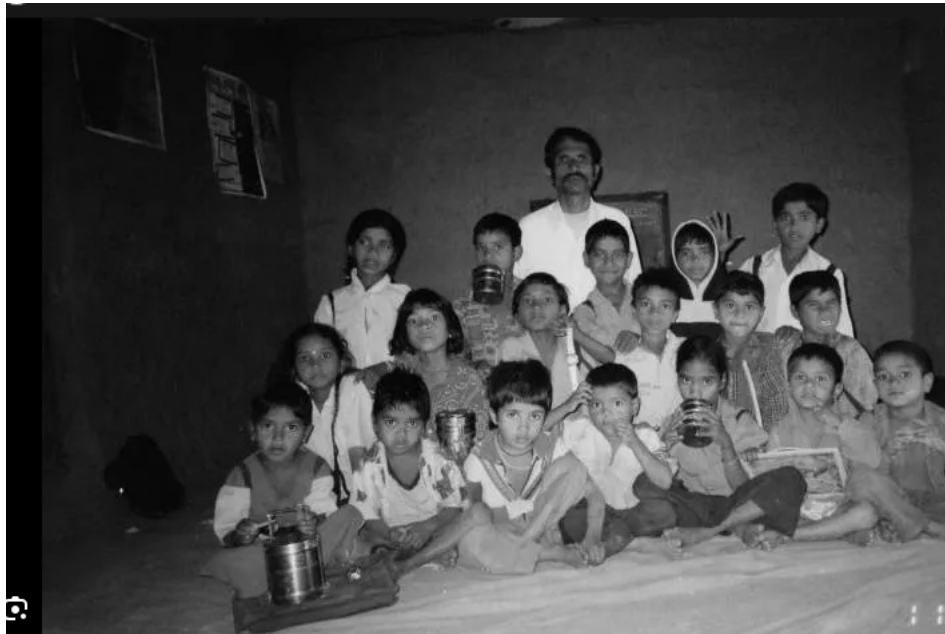


TABLE 2—TEACHER ATTENDANCE

September 2003–February 2006			Difference between treatment and control schools		
Treatment (1)	Control (2)	Diff (3)	Until mid-test (4)	Mid- to post-test (5)	After post-test (6)
<i>Panel A. All teachers</i>					
0.79	0.58	0.21 (0.03)	0.20 (0.04)	0.17 (0.04)	0.23 (0.04)
1,575	1,496	3,071	882	660	1,529
<i>Panel B. Teachers with above median test scores</i>					
0.78	0.63	0.15 (0.04)	0.15 (0.05)	0.15 (0.05)	0.14 (0.06)
843	702	1,545	423	327	795
<i>Panel C. Teachers with below median test scores</i>					
0.78	0.53	0.24 (0.04)	0.21 (0.05)	0.14 (0.06)	0.32 (0.06)
625	757	1,382	412	300	670

Notes: Child learning levels were assessed in a mid-test (April 2004) and a post-test (November 2004). After the post-test, the “official” evaluation period was ended. Random checks continued in both the treatment and control schools. Standard errors are clustered by school. Panels B and C only include the 109 schools where teacher tests were available.

TABLE 9—ESTIMATION OF TREATMENT EFFECTS FOR THE MID- AND POST-TEST

Mid-test				Post-test			
Took written (1)	Math (2)	Lang. (3)	Total (4)	Took written (5)	Math (6)	Lang. (7)	Total (8)
<i>Panel A. All children</i>							
0.04 (0.03)	0.15 (0.07)	0.16 (0.06)	0.17 (0.06)	0.06 (0.04)	0.21 (0.12)	0.16 (0.08)	0.17 (0.09)
1,893	1,893	1,893	1,893	1,760	1,760	1,760	1,760
<i>Panel B. With controls</i>							
0.04 (0.03)	0.13 (0.07)	0.14 (0.06)	0.14 (0.06)	0.06 (0.04)	0.18 0.13	0.14 0.08	0.15 0.09
1,752	1,752	1,752	1,752	1,760	1,760	1,760	1,760
<i>Panel C. Took pretest oral</i>							
	0.14 (0.08)	0.13 (0.06)	0.15 (0.07)		0.2 (0.14)	0.13 (0.09)	0.16 (0.10)
	1,550	1,550	1,550		1,454	1,454	1,454
<i>Panel D. Took pretest written</i>							
	0.19 (0.12)	0.28 (0.11)	0.25 (0.11)		0.28 (0.18)	0.28 (0.11)	0.25 (0.12)
	343	343	343		306	306	306
<i>Panel E. Girls</i>							
0.07 (0.03)	0.18 (0.07)	0.18 (0.07)	0.19 (0.07)	0.07 (0.05)	0.22 (0.12)	0.13 0.08	0.17 (0.09)
876	876	876	876	811	811	811	811
<i>Panel F. Boys</i>							
0.02 (0.04)	0.11 (0.09)	0.13 (0.07)	0.13 (0.07)	0.05 (0.04)	0.20 (0.15)	0.15 (0.10)	0.16 (0.10)
983	983	983	983	926	926	926	926

Notes: The table presents the coefficient estimate of being in a treated school on the sum of a child's score on the oral and written exams. All regressions include controls for the child's learning levels prior to the program. The mid- and post-test scores are normalized by mid-test control group. Controls in panel B include Block, Teacher Test Scores, and Infrastructure Index. Standard errors are clustered by school.

Going Beyond the RCT

- DHR want to understand the mechanism underlying their effects and whether a more effective incentive scheme could be found.
- Hard to answer these questions with reduced form evidence → need a structural model!
- Key incentive structure of the intervention:
 - If a teacher has already attended 10 days with a month, she has a strong incentive to continue attending the remaining days until the end of the month (“teacher is in the money”).
 - If a teacher has attended so few days that she cannot reach ten, she has very little incentives to attend (“teacher is out of money”)
 - Everything resets at the beginning of the month, teachers who were in the money are not longer so and at the beginning of the month there is only a dynamic incentive to attend because that might lead teachers to be in the money.

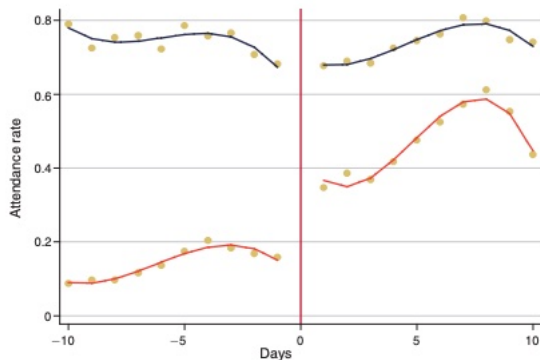


FIGURE 3. RDD REPRESENTATION OF TEACHER ATTENDANCE AT THE START AND END OF THE MONTH

Notes: The top lines represent the months in which the teacher is in the money, while the bottom lines represent the months in which the teacher is not in the money. The estimation includes a third-order polynomial of days on the left and right side of the change of month.

Dynamic Labor Supply model

- Let m be the month and t the day within the month, where $t = \{1, \dots, T_m\}$.
- Let U_{tm} be the flow utility of a teacher on day t of month m . This utility function is assumed to be separable in consumption and leisure and takes the following form

$$U_{tm} = \beta C_{tm}(\pi) + \mu_{tm} L_{tm} \quad (16)$$

- where $C_{tm}(\pi)$ is consumption in day t month m which is a function of total monthly income π . β converts consumption measured in Rupees into utility. L_{tm} is a binary variable which is one if the teacher does not go to work on day t and μ_{tm} measures the value of leisure which takes the following structure

$$\mu_{tm} = \mu + \epsilon_{tm} \quad (17)$$

where $\mu \sim \mathcal{N}(\mu_1; \sigma_1)$ and ϵ_{tm} is iid standard normal.

- Utility is linear in consumption and consumption and leisure are separable, there is not going to be a dependency in behavior across months.
- DHR assume no discounting within months and since utility is linear in consumption, in this model teacher consumes all her income on the last day of the month, when she is paid.
- Incentive to work

$$\pi(d_{T_m}) = 500 + 50 \max(0, d_{T_m} - 10) \quad (18)$$

where d_{T_m} represents how many days of work an agent has accumulated in a given month. In the control group, π_m is fixed at 1000 regardless of the accumulated days of work.

Dynamic Labor Supply model

- The dynamic problem for a teacher in a particular month m can be written as follows, for $t < T_m$

$$V_m(t, d_{t-1}; \mu_{tm}) = \max \{ \mu_{tm} + E[V_m(t+1, d_{t-1}; \mu_{t+1m})], E[V_m(t+1, d_{t-1}+1; \mu_{t+1m})] \} \quad (19)$$

d_{t-1} denotes how many days of work a particular teacher has accumulated when entering period t .

- From (19), we can see that teachers trade off immediate gratification from leisure against the possibility of higher wages at the end of the month. When $t = T_m$,

$$V_m(T_m, d_{T_m-1}; \mu_{T_m}) = \max \{ \mu_{T_m} + \beta \pi(d_{T_m-1}), \beta \pi(d_{T_m-1} + 1) \} \quad (20)$$

note that given the assumptions of the model, choices are independent across months.

- DHR assumes that μ is drawn *independently* from a standard normal distribution with mean μ_1 and variance σ_1 . The distribution μ is drawn anew each month. Under this assumption,

$$\Pr(\text{work}_{T_m} = 1 | D_{T_m-1} = d_{T_m-1}; \mu_1, \sigma_1) = \int \Phi(\beta\{\pi(d_{T_m-1} + 1) - \pi(d_{T_m-1})\} - (\mu_1 + \sigma_1 u_s)) \phi\left(\frac{u_s - \mu_1}{\sigma_1}\right) du_s$$

- This is SML! DHR uses this model to fit the model on the treatment group. It then verifies whether the model is correct by checking the predictions of the model when setting pay equal to 1000 Rps (regardless of days worked) and benchmark the results with what is observed in the control group.
- After choosing the model with the best fit, DHR play around with different incentives structure

TABLE 5—COUNTERFACTUAL COST-MINIMIZING POLICIES

Expected days worked (1)	Bonus cutoff (2)	Bonus (3)	Expected cost (4)	Test score gain over control group (13 days) (5)
14	0	0	500	0.04
15	21	25	521	0.07
16	22	75	664	0.11
17	21	75	672	0.15
18	20	75	755	0.18
19	20	100	921	0.22
20	20	125	1,112	0.26
21	16	225	2,642	0.29
22	11	275	4,604	0.33

Appendix

Proof

For any ε we have with probability approaching 1:

- a) $\hat{Q}_N(\hat{\theta}) > \hat{Q}_N(\theta_0) - \varepsilon$ (by virtue of maximizer)
- b) $Q_0(\hat{\theta}) > \hat{Q}_N(\hat{\theta}) - \varepsilon$ (by uniform convergence)
- c) $\hat{Q}_N(\theta_0) > Q_0(\theta_0) - \varepsilon$ (by uniform convergence)

Therefore:

$$Q_0(\hat{\theta}) \stackrel{b)}{>} \hat{Q}_N(\hat{\theta}) - \varepsilon \stackrel{a)}{>} \hat{Q}_N(\theta_0) - 2\varepsilon \stackrel{c)}{>} Q_0(\theta_0) - 3\varepsilon.$$

Proof

For any ε we have with probability approaching 1:

- a) $\hat{Q}_N(\hat{\theta}) > \hat{Q}_N(\theta_0) - \varepsilon$ (by virtue of maximizer)
- b) $Q_0(\hat{\theta}) > \hat{Q}_N(\hat{\theta}) - \varepsilon$ (by uniform convergence)
- c) $\hat{Q}_N(\theta_0) > Q_0(\theta_0) - \varepsilon$ (by uniform convergence)

Therefore:

$$Q_0(\hat{\theta}) \stackrel{b)}{>} \hat{Q}_N(\hat{\theta}) - \varepsilon \stackrel{a)}{>} \hat{Q}_N(\theta_0) - 2\varepsilon \stackrel{c)}{>} Q_0(\theta_0) - 3\varepsilon.$$

Let $\mathcal{N}$ be an open neighborhood of θ_0 and $\mathcal{N}^c$ its complement.

Note that $\sup_{\theta \in \Theta \cap \mathcal{N}^c} Q_0(\theta) = Q_0(\theta^*) < Q_0(\theta_0)$.

- Choose $3\varepsilon = Q_0(\theta_0) - \sup_{\theta \in \Theta \cap \mathcal{N}^c} Q_0(\theta)$.
- It follows that $Q_0(\hat{\theta}) > \sup_{\theta \in \Theta \cap \mathcal{N}^c} Q_0(\theta)$ w.p.a.1.

Hence $\hat{\theta} \in \mathcal{N}$ $\square$

Proof Sketch

Start w/ FOC:

$$\nabla_{\theta} \hat{Q}_N(\hat{\theta}) = 0$$

- Mean value expansion:

$$\nabla_{\theta} \hat{Q}_N(\hat{\theta}) = \nabla_{\theta} \hat{Q}_N(\theta_0) + \hat{\mathbf{H}}(\bar{\theta})(\hat{\theta} - \theta_0)$$

where $\bar{\theta} \in [\hat{\theta}, \theta_0]$ and $\hat{\mathbf{H}}(\theta) \equiv \nabla_{\theta\theta} \hat{Q}_N(\theta)$.

- Rearranging we get:

$$\sqrt{N}(\hat{\theta} - \theta_0) = -\hat{\mathbf{H}}(\bar{\theta})^{-1} \left(\sqrt{N} \nabla_{\theta} \hat{Q}_N(\theta_0) \right)$$

Proof Sketch

Start w/ FOC:

$$\nabla_{\theta} \hat{Q}_N(\hat{\theta}) = 0$$

- Mean value expansion:

$$\nabla_{\theta} \hat{Q}_N(\hat{\theta}) = \nabla_{\theta} \hat{Q}_N(\theta_0) + \hat{\mathbf{H}}(\bar{\theta})(\hat{\theta} - \theta_0)$$

where $\bar{\theta} \in [\hat{\theta}, \theta_0]$ and $\hat{\mathbf{H}}(\theta) \equiv \nabla_{\theta\theta} \hat{Q}_N(\theta)$.

- Rearranging we get:

$$\sqrt{N}(\hat{\theta} - \theta_0) = -\hat{\mathbf{H}}(\bar{\theta})^{-1} \left(\sqrt{N} \nabla_{\theta} \hat{Q}_N(\theta_0) \right)$$

- Want to show:

$$\sqrt{N}(\hat{\theta} - \theta_0) = -\mathbf{H}(\theta_0)^{-1} \left(\sqrt{N} \nabla_{\theta} \hat{Q}_N(\theta_0) \right) + o_p(1)$$

Proof Sketch

Start w/ FOC:

$$\nabla_{\theta} \hat{Q}_N(\hat{\theta}) = 0$$

- Mean value expansion:

$$\nabla_{\theta} \hat{Q}_N(\hat{\theta}) = \nabla_{\theta} \hat{Q}_N(\theta_0) + \hat{\mathbf{H}}(\bar{\theta})(\hat{\theta} - \theta_0)$$

where $\bar{\theta} \in [\hat{\theta}, \theta_0]$ and $\hat{\mathbf{H}}(\theta) \equiv \nabla_{\theta\theta} \hat{Q}_N(\theta)$.

- Rearranging we get:

$$\sqrt{N}(\hat{\theta} - \theta_0) = -\hat{\mathbf{H}}(\bar{\theta})^{-1} \left(\sqrt{N} \nabla_{\theta} \hat{Q}_N(\theta_0) \right)$$

- Want to show:

$$\sqrt{N}(\hat{\theta} - \theta_0) = -\mathbf{H}(\theta_0)^{-1} \left(\sqrt{N} \nabla_{\theta} \hat{Q}_N(\theta_0) \right) + o_p(1)$$

- Need $\left\| \hat{\mathbf{H}}(\bar{\theta}) - \mathbf{H}(\theta_0) \right\| \xrightarrow{p} 0$

Proof Sketch

Triangle inequality:

$$\begin{aligned}\left\| \hat{\mathbf{H}}(\bar{\theta}) - \mathbf{H}(\theta_0) \right\| &\leq \underbrace{\left\| \hat{\mathbf{H}}(\bar{\theta}) - \mathbf{H}(\bar{\theta}) \right\|}_{\text{Noise in Hessian}} + \underbrace{\left\| \mathbf{H}(\bar{\theta}) - \mathbf{H}(\theta_0) \right\|}_{\text{Noise in parameter}} \\ &\leq \sup_{\theta \in \Theta} \left\| \hat{\mathbf{H}}(\theta) - \mathbf{H}(\theta) \right\| + \left\| \mathbf{H}(\bar{\theta}) - \mathbf{H}(\theta_0) \right\|\end{aligned}$$

Proof Sketch

Triangle inequality:

$$\begin{aligned}\left\| \hat{\mathbf{H}}(\bar{\theta}) - \mathbf{H}(\theta_0) \right\| &\leq \underbrace{\left\| \hat{\mathbf{H}}(\bar{\theta}) - \mathbf{H}(\bar{\theta}) \right\|}_{\text{Noise in Hessian}} + \underbrace{\left\| \mathbf{H}(\bar{\theta}) - \mathbf{H}(\theta_0) \right\|}_{\text{Noise in parameter}} \\ &\leq \sup_{\theta \in \Theta} \left\| \hat{\mathbf{H}}(\theta) - \mathbf{H}(\theta) \right\| + \left\| \mathbf{H}(\bar{\theta}) - \mathbf{H}(\theta_0) \right\|\end{aligned}$$

- By assumption $\sup_{\theta \in \Theta} \left\| \hat{\mathbf{H}}(\theta) - \mathbf{H}(\theta) \right\| \xrightarrow{P} 0$ (kills 1st term)
- Because $\hat{\theta} \xrightarrow{P} \theta_0$, it must also be that $\bar{\theta} \xrightarrow{P} \theta_0$ (squeeze theorem).
- Since $\mathbf{H}(\cdot)$ is continuous, it follows from the continuous mapping theorem that $\left\| \mathbf{H}(\bar{\theta}) - \mathbf{H}(\theta_0) \right\| \xrightarrow{P} 0$ (kills 2nd term)
- Therefore $\hat{\mathbf{H}}(\bar{\theta}) \xrightarrow{P} \mathbf{H}(\theta_0)$

Recall

$$\sqrt{N} \left(\hat{\theta} - \theta_0 \right) = -\hat{\mathbf{H}} \left(\bar{\theta} \right)^{-1} \left(\sqrt{N} \nabla_{\theta} \hat{Q}_N \left(\theta_0 \right) \right)$$

Recall

$$\sqrt{N}(\hat{\theta} - \theta_0) = -\hat{\mathbf{H}}(\bar{\theta})^{-1} \left(\sqrt{N} \nabla_{\theta} \hat{Q}_N(\theta_0) \right)$$

- We established $\hat{\mathbf{H}}(\bar{\theta}) \xrightarrow{P} \mathbf{H}(\theta_0)$
- By continuous mapping theorem, $\hat{\mathbf{H}}(\bar{\theta})^{-1} \xrightarrow{P} \mathbf{H}(\theta_0)^{-1}$
- And by assumption: $\sqrt{N} \nabla_{\theta} \hat{Q}_N(\theta_0) \xrightarrow{d} N(0, \mathbf{V})$
- Therefore, by Slutsky,
$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0, \mathbf{H}(\theta_0)^{-1} \mathbf{V} \mathbf{H}(\theta_0)^{-1}\right) \quad \square$$

Example: “folding tent”

$$Q_N(\theta) = \begin{cases} \max\{4N\theta - 2, 0\} & \text{if } \theta \leq \frac{3}{4N} \\ \max\{-4N\theta + 4, 0\} & \text{if } \theta > \frac{3}{4N} \end{cases}$$

Example: “folding tent”

$$Q_N(\theta) = \begin{cases} \max\{4N\theta - 2, 0\} & \text{if } \theta \leq \frac{3}{4N} \\ \max\{-4N\theta + 4, 0\} & \text{if } \theta > \frac{3}{4N} \end{cases}$$

Example: “folding tent”

$$Q_N(\theta) = \begin{cases} \max\{4N\theta - 2, 0\} & \text{if } \theta \leq \frac{3}{4N} \\ \max\{-4N\theta + 4, 0\} & \text{if } \theta > \frac{3}{4N} \end{cases}$$

- $\Theta = [0, 1]$ is compact

Example: “folding tent”

$$Q_N(\theta) = \begin{cases} \max\{4N\theta - 2, 0\} & \text{if } \theta \leq \frac{3}{4N} \\ \max\{-4N\theta + 4, 0\} & \text{if } \theta > \frac{3}{4N} \end{cases}$$

- $\Theta = [0, 1]$ is compact
- $Q_N(\cdot)$ is continuous

Example: “folding tent”

$$Q_N(\theta) = \begin{cases} \max\{4N\theta - 2, 0\} & \text{if } \theta \leq \frac{3}{4N} \\ \max\{-4N\theta + 4, 0\} & \text{if } \theta > \frac{3}{4N} \end{cases}$$

- $\Theta = [0, 1]$ is compact
- $Q_N(\cdot)$ is continuous
- $\lim_{N \rightarrow \infty} Q_N(\theta) = 0 \quad \forall \theta \in [0, 1]$ (pointwise convergence)

Example: “folding tent”

$$Q_N(\theta) = \begin{cases} \max\{4N\theta - 2, 0\} & \text{if } \theta \leq \frac{3}{4N} \\ \max\{-4N\theta + 4, 0\} & \text{if } \theta > \frac{3}{4N} \end{cases}$$

- $\Theta = [0, 1]$ is compact
- $Q_N(\cdot)$ is continuous
- $\lim_{N \rightarrow \infty} Q_N(\theta) = 0 \quad \forall \theta \in [0, 1]$ (pointwise convergence)
- But $\max_{\theta \in \Theta} |Q_N(\theta) - 0| = 1 \quad \forall N \in \mathbb{N}!$